🌀 M003 Extension — The Loop Spiral (Beginner)

Topic: One loop that **turns** (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after the Loop Staircase) · **Time:** about 30 minutes

The **staircase** went in a **straight** line. A **spiral** is different — every pass it **turns**. Same loop, new direction each time. Type `SPIRAL` in the chat and watch it **climb** and **turn**.

```
x = 8
y = 0
z = 8
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    if step % 4 == 0:
        x = x + 4
    elif step % 4 == 1:
        z = z + 4
    elif step % 4 == 2:
        x = x - 4
    else:
        z = z - 4
```

> 🔁 **Big idea:** `step % 4` counts **0, 1, 2, 3, 0, 1, 2, 3 …** — four directions on repeat. That is the **turn**. `y = y + 1` lifts each square higher.

	3 ↑ back **z − 4**	
2 ← left **x − 4**	**step % 4**	**0 → right** **x + 4**
	1 ↓ forward **z + 4**	

1 · Predict 🔮

Read the code. Circle one.

`range(12)` **— how many squares does the loop build? Circle one:** 4 · 8 · 12

`step % 4` **gives which numbers? Circle one:** 0 1 2 3 · 1 2 3 4 · 0 1 2

When `step % 4` **is** 0**, which way does it go? Circle one:** right · up · back

Which line makes the spiral go UP? Circle one: `y = y + 1` · `x = x + 4` · `step % 4`

2 · Spot the Bug 🐛

Bug A — no turn. A friend deleted the four `if` lines, so the direction never changes.

```
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    x = x + 4
```

What do you build now? Circle one: a straight diagonal · a spiral · nothing

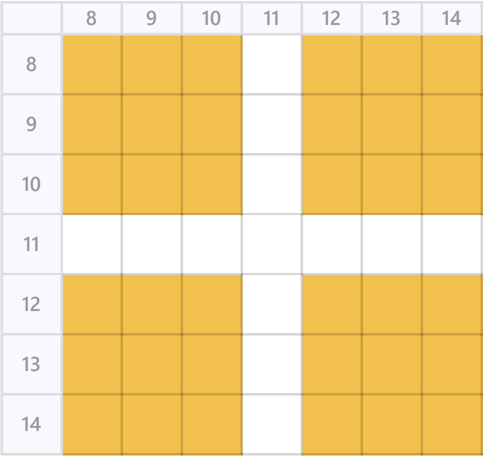
Bug B — no rise. One line is missing, so the tower stays flat on the floor.

Which line is missing? Circle one: `y = y + 1` · `x = x + 4` · `z = z - 4`

3 · Show Your Work 📸 🎥

Load the world. Stand on your home spot. Type `SPIRAL` and watch each square **turn** to a new corner and **climb** one block.

👁 Top-down view (looking DOWN): x → across, z ↓ deeper



Looking down, the squares sit on 4 corners — the loop turns right, forward, left, back, then repeats, climbing each time.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does.

2 · Your build. Point the camera at the spiral. Show it climbing and turning.

Say these out loud, filling in the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.